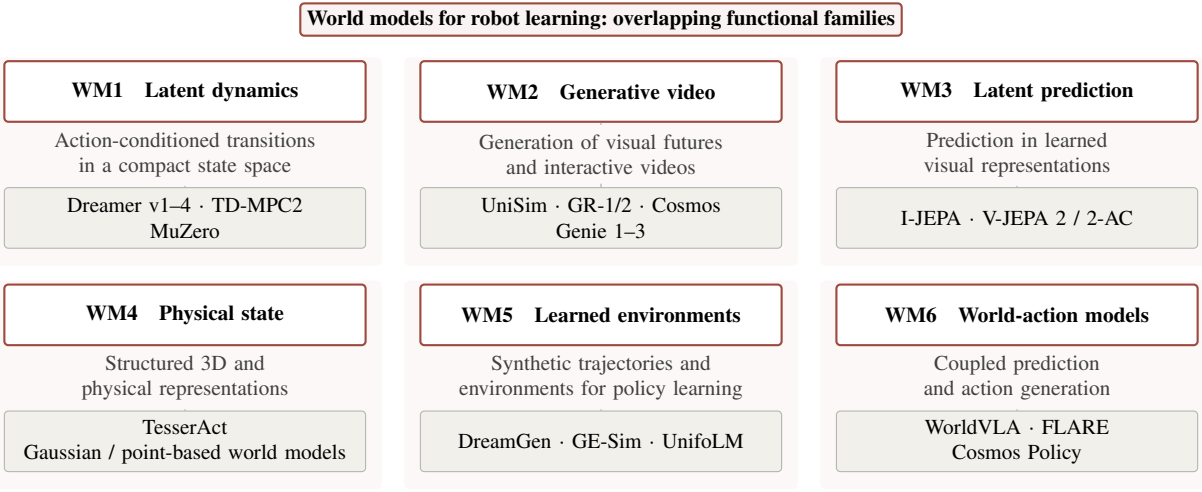
World models for robot learning: overlapping functional families
WM1 Latent dynamics
Action-conditioned transitions in a compact state space
Dreamer v1–4 · TD-MPC2
MuZero
WM2 Generative video
Generation of visual futures and interactive videos
UniSim · GR-1/2 · Cosmos
Genie 1–3
WM3 Latent prediction
Prediction in learned visual representations
I-JEPA · V-JEPA 2 / 2-AC
WM4 Physical state
Structured 3D and physical representations
TesserAct
Gaussian / point-based world models
WM5 Learned environments
Synthetic trajectories and environments for policy learning
DreamGen · GE-Sim · UnifoLM
WM6 World-action models
Coupled prediction and action generation
WorldVLA · FLARE
Cosmos Policy
Families describe model function; placement does not establish a remedy for a VLA limitation.